BEATS: Audio Pre-Training with Acoustic Tokenizers

Sanyuan Chen, Yu Wu, Chengyi Wang, Shujie Liu, Daniel Tompkins, Zhuo Chen, Wanxiang Che, Xiangzhan Yu, Furu Wei

Harbin Institute of Technology, Harbin, Heilongjiang, China
Microsoft Research Asia, Beijing, China
Nankai University, Tianjin, China
Microsoft Corporation, Redmond, WA, USA

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Why Audio SSL Matters

- Audio powers search, safety, accessibility, and ambient intelligence.
- SSL enables learning from massive unlabeled audio across domains.
- Unified audio representations unlock transfer to many downstream tasks.

Why General Audio Is Hard

- Audio is diverse: speech, music, events, overlapping sources, noise.
- Existing SSL often reconstructs low-level spectrogram details.
- We need semantics-focused pre-training robust to nuisance factors.

Limitations of Reconstruction-Based Audio SSL

- Optimizes pixel/TF fidelity, not event semantics.
- Sensitive to reverberation, channel, and noise perturbations.
- Inefficient: reconstructs masked content at high resolution.
- Lags behind discrete prediction paradigms in vision/speech/NLP.

Core Idea & Contributions Overview

- Replace reconstruction with masked discrete label prediction for audio.
- Iteratively co-train an acoustic tokenizer and an audio SSL model.
- Design tokenizers that produce semantic, noise-robust labels.
- Efficient ViT-based Masked Audio Modeling with 75% masking.
- SOTA on AudioSet and ESC-50 with fewer parameters, no external data.

Method Overview: Iterative BEATS Framework

- Tokenizer labels audio patches with discrete codes.
- SSL model predicts masked labels from unmasked context.
- Trained SSL model distills a better tokenizer; repeat.

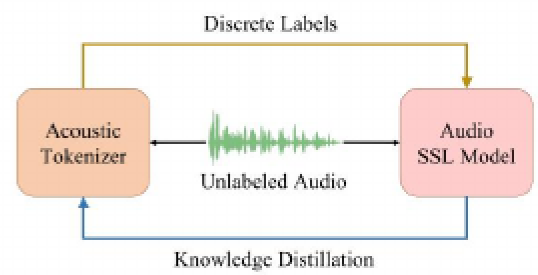

Figure: Iterative audio pre-training of BEATS.

Acoustic Tokenizers: Two Designs

- Random-Projection Tokenizer: cold-start labels via nearest codebook.
- Self-Distilled Tokenizer: Transformer + VQ with learnable codebook.
- Distillation from SSL or supervised teacher injects semantics.

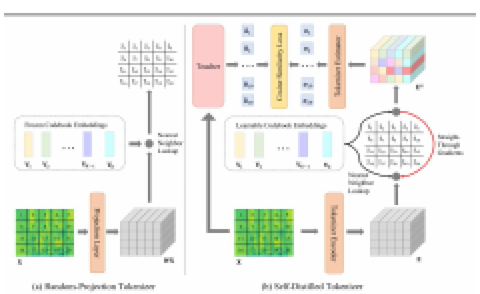

Figure: Acoustic tokenizers for discrete label generation.

Random-Projection Tokenizer (Cold Start)

- Random linear projection maps patches to embedding space.
- Fixed codebook assigns nearest index as discrete label.
- Simple, fast, and sufficient to bootstrap BEATSiter1.

Self-Distilled Tokenizer (SDT)

- Transformer encoder produces patch embeddings.
- Vector Quantization assigns codes; EMA-updated codebook.
- Estimator learns to match teacher outputs (cosine loss).
- Teacher can be SSL-pretrained or supervised fine-tuned.

Masked Audio Modeling (MAM) Objective

- Mask 75% of patches; encode only unmasked for efficiency.
- Predict tokenizer labels at masked positions (cross-entropy).
- ViT backbone with relative positions and DeepNorm.

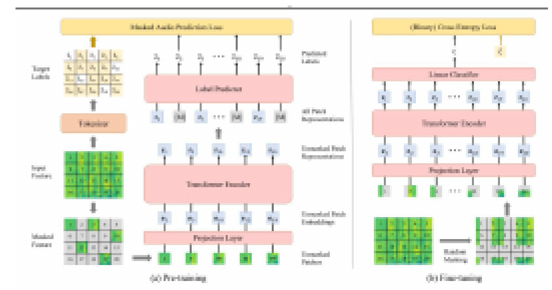

Figure: Overview of audio SSL model pre-training and fine-tuning.

Efficiency by Design

- Feed only unmasked patches into the encoder.
- Heavy masking boosts speed and regularization.
- Discrete targets simplify the prediction head.

Fine-Tuning Protocol for Downstream Tasks

- Discard label predictor; keep encoder as backbone.
- Mean-pool features; add linear head (CE/BCE).
- Use SpecAug and mixup where appropriate.

Iterative Training and EM-Style Convergence

- Alternate improving labels (tokenizer) and model (SSL).
- Formulated as EM-like optimization over latent labels.
- Guarantees non-decreasing data likelihood per iteration.

Experimental Setup: Datasets & Metrics

- Pre-train on AudioSet-2M; evaluate on AS-2M/20K (mAP).
- Transfer to ESC-50, Speech Commands (V1/V2), IEMOCAP ER (acc).
- Model: ViT Base ($\tilde{9}$0M) with relative positions and DeepNorm.

Baselines and Training Details

- Compare to strong supervised and SSL baselines (e.g., ViT-based and MAE-style).
- No external labeled data for BEATS; fair transfer protocols.
- Strong augmentation and class balancing per task when needed.

Single-Model Results

- AS-2M mAP 48.6 with a Base ViT, no external data
- AS-20K mAP 38.9
- ESC-50 accuracy 98.1%
- BEATSiter1 already beats reconstruction SSL; self-distilled tokenizers add gains

Robustness, Low-Data, and Transfer Insights

- Tokens emphasize stable cues over nuisance variations.
- Iterative distillation improves low-resource performance.
- Unified objective simplifies cross-task transfer pipelines.

Practical Considerations and Limitations

- Iteration adds compute overhead (tokenizer + model loops).
- Reliance on AudioSet; broader corpora may help coverage.
- Model size is modest vs frontier; scaling likely beneficial.

Acknowledgments

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Open Questions

- How to scale tokenizers without labels?
- Can tokens unify cross-domain audio tasks?
- How to co-train audio with vision/language tokens?
- What guarantees beyond EM-style analysis?
- How do tokens evolve with model/data scale?